

Workshop Problem

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Problem 0.

Let $r(t) = \langle 2\sin(t), 2\cos(t), t \rangle$ where $t > 0$ be the vector function that describes the position of a fly (imagine this as spiral with the center as the z axis). t represents time in seconds, and let's assume the position is measured in meters.

Q1: What is the speed of the fly at $t = \frac{\pi}{2}$ seconds?

We know the velocity vector is the derivative of the position vector so $v(t) = r'(t)$. Thus, $v(t) = \langle 2\cos(t), -2\sin(t), 1 \rangle$ The velocity vector at $t = \frac{\pi}{2}$ is $v(\frac{\pi}{2}) = \langle 0, -2, 1 \rangle$ The speed is the magnitude of velocity. So, speed at $t = \frac{\pi}{2}$ is $|\langle 0, -2, 1 \rangle| = \sqrt{5}m/s$

Q2: What is the acceleration of the fly at $t = \frac{\pi}{2}$ seconds?

We know the acceleration vector is the derivative of the velocity vector so $a(t) = v'(t)$. Thus, $a(t) = \langle -2\sin(t), -2\cos(t), 0 \rangle$ The acceleration vector at $t = \frac{\pi}{2}$ is $a(\frac{\pi}{2}) = \langle -2, 0, 0 \rangle$ The acceleration magnitude is $\sqrt{4} = 2m/s^2$ Visualizing this, we can say that the fly experiences an acceleration of $2m/s^2$ in the -x direction.

Q3: Is the fly slowing down, speeding up, or neither at $t = \frac{\pi}{2}$ seconds?

If you have done any previous calculus courses that deal with a similar problem in 2-dimensions, you know an object will be speeding up when its velocity and acceleration is in the same direction. The object will slow down when its velocity and acceleration is in the opposite direction. We can check if the acceleration and velocity is in the same direction by using the dot product. If the dot product between the acceleration vector and the velocity vector is positive (> 0), then we know that there is a component of the acceleration vector that is in the same direction as the velocity vector. If the dot product is negative (< 0), then there is a component of the acceleration vector that is in the opposite direction as the velocity vector.

So we find the dot product $v(\frac{\pi}{2}) \cdot a(\frac{\pi}{2}) = \langle 0, -2, 1 \rangle \cdot \langle -2, 0, 0 \rangle = 0 + 0 + 0 = 0$. We see that the dot product is equal to zero. This is a rather interesting solution because that means the acceleration and velocity vector is perpendicular! That means the object is neither speeding up nor slowing down, it is moving at a constant speed. If you have taken any physics courses before, this is exactly analogous to centripetal force and uniform circular motion. You should know that the centripetal force in uniform circular motion is always perpendicular to the velocity of the object!

Q4: At what time does the fly intersect the $z=4$ plane?

Remember that $r(t) = \langle 2\sin(t), 2\cos(t), t \rangle$. A very straightforward answer is that $t = 4$ seconds.

Q5: You get annoyed and throw a rock at the fly. The rock travels in a line path represented by the vector equation $r_1(t) = \langle 2t, 4t, 2t \rangle$ (treat the rock as a point in space). Does your rock ever hit the fly?

Remember that $r(t) = \langle 2\sin(t), 2\cos(t), t \rangle$. We find if the two vector functions intersect. So there should be a t that satisfies $r(t) = r_1(t)$.

$$2\sin(t) = 2t \tag{1}$$

$$2\cos(t) = 4t \tag{2}$$

$$t = 2t \tag{3}$$

From (3) we get that $t = 0$ seconds. Then we check with equation (2) .

$$2(\cos(0)) = 4(0) \tag{4}$$

$$2 * (1) = 0 \tag{5}$$

$$2 \neq 0 \tag{6}$$

Equation (6) is a contradiction. So the two vector functions never intersect. You have bad aim and missed the fly.

Challenge Problem! At what time t is the rock closest to the fly?

This problem will require you to find a function of the distance between the rock and the fly, then take the derivative of it with respect to time. There will be lot of a chain rule and product rule, but the answer should be solvable. If you figure it out email me. *Hint:* Can you represent the distance with a vector?